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Magnetostratigraphy and timing of the Oligocene Ethiopian traps

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Abstract

A combined paleomagnetic, $^{40}Ar/^{39}Ar$ and geochemical investigation has been conducted in two type sections (65 sites) of the NW Ethiopian plateau volcanic pile. The main 2 km-thick complete section of the flood basalts at Lima-Limo contains a succession of only three magnetic chrons. The central normal chron appears to correspond to Chron C11n, according to the $^{40}Ar/^{39}Ar$ ages, which cluster around 30 Ma. Two alternative magnetostratigraphic interpretations both indicate a duration on the order of or less than 1 Myr. A detailed major element study of the section shows remarkable magma homogeneity, with two cycles of differentiation. Using correlation with other sections, we propose that the emplacement of the bulk of the Ethiopian igneous province (with an estimated volume on the order of 10^6 km^3) in about 1 Myr or less should be linked with the Oi2 global cooling event, which occurs in Chron C11r. Further independent support for this identification comes from the finding of prominent tephra horizons in central Indian Ocean sediments from leg 115, with a biostratigraphic age coinciding with the Oi2 event. The 30 Ma African pole produced by this study (77ºN, 208ºE, $A_{95} = 3.7$º, $N = 53$) is far-sided by $6º \pm 6º$ with respect to the reference synthetic pole for Africa and yields a large paleosecular variation. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Ar-40/Ar-39; Oligocene; Ethiopia; flood basalts; magnetostratigraphy

1. Introduction and geological context

Traps constitute the most voluminous continental volcanic provinces in the geological record and have thus attracted a lot of attention. Determination of their precise age and duration has been the focus of several recent studies. For that purpose, the combined use of $^{40}Ar/^{39}Ar$ geochronology and magnetostratigraphy has proved to be very efficient. For the Deccan, Parana and Siberian traps, this approach has indicated that the bulk of the lava pile was emplaced in about 1 Myr or less [1,2]. Besides their interest in terms of mantle dynamics, plume birth and continental break-up [3,4], these chronological constraints may have major significance for the geological timescale.

The magmatic flux during trap emplacement has the potential to induce global climatic change, through the massive input of ashes and gases in the up-

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per stratosphere [5]. These considerations have led to the hypothesis that many, if not all, major climatic and biological crises are synchronous with trap emplacement [1]. This seems to be the case for the Deccan, Parana and Siberian traps, which are synchronous with the Cretaceous–Paleogene (K/T), Jurassic–Cretaceous and Permian–Triassic (P/T) boundaries respectively. This 'volcanic' thesis is challenged by an 'extraterrestrial' thesis, which would correlate major boundaries with large asteroid impacts. The latter appears to be vindicated for the K/T boundary (e.g. [6]), but the Deccan traps are essentially contemporaneous [1,7] giving rise to the possibility that both types of geological catastrophe occurred at that time. The debate may arise again for a geological boundary of lesser significance than K/T or P/T, namely the Eocene–Oligocene (E/O) boundary. Flood basalt volcanism in the Ethiopian traps [2,8], an iridium anomaly observed in Italy [9] and two large impact craters recently dated at 35.5 Ma [10] all seem to be found in a period range close to the E/O boundary, and to the bioclimatic changes that occurred over a protracted period on either side of that boundary [11].

The Paleogene Ethiopian volcanic province [12] was not investigated using the above mentioned techniques until a French–Ethiopian cooperative project was launched in 1993. The province is now split by continental break-up around the Afar depression, with the major part forming the NW Ethiopian plateau (partly dismantled to the N in Eritrea), and minor parts on the east side of the main Ethiopian rift (Somalia plate) and on the Arabian plate in Yemen (Fig. 1). The southern Ethiopian Eocene volcanic province, which may constitute a separate province both in time and space [13], is not included in the present discussion.

From former studies dating back to the seventies, only K/Ar ages were available, ranging from 15 to 40 Ma [12,14], and regional paleomagnetic sampling could not be used for stratigraphic purposes [15,16] (see overview in [17]). The Ethiopian traps display two singular features with respect to older traps: it includes an important acidic component (up to 400 m within a total of about 2000 m in the central part), and it is overlain to a great extent by younger volcanic series.

Our first results [2,8] on the NW Ethiopian plateau were based on field sampling undertaken

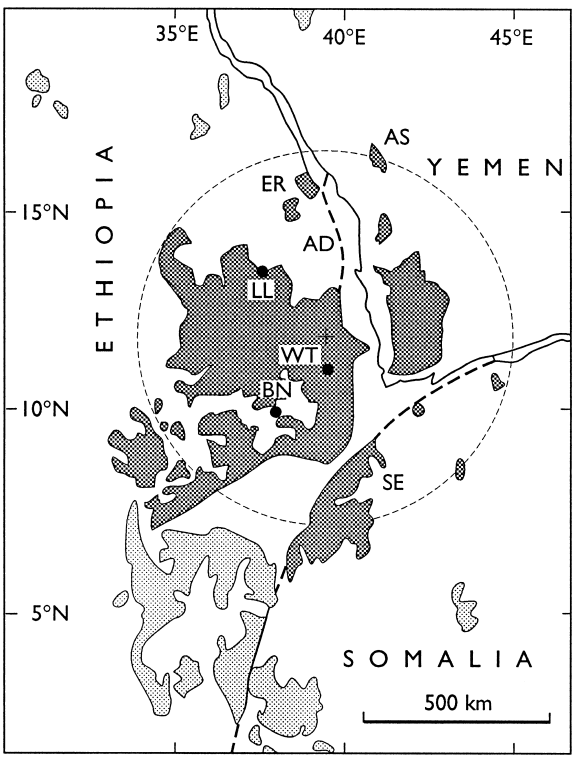


Fig. 1. Sketch map of the Oligocene Ethiopian trap Formation (in dark gray, light gray when identification is questionable), with the two studied sections Lima-Limo (*LL*) and Wegel Tena (*WT*), as well as location of other mentioned areas: Blue Nile, Adigrat, Eritrea, As Sarat volcanics, SE province (respectively *BN*, *AD*, *ER*, *AS*, *SE*). A circle of 500 km radius enclosing most remnants (except to the south) is proposed as an estimate of initial trap extent at 30 Ma. Somalia and Arabia have been rotated in their initial position according to [25].

in 1993–94: we presented what we felt was a strong case for rapid emplacement of the whole lava pile. All robust $^{40}Ar/^{39}Ar$ ages fall in the range 30 ± 1 Ma, while a selection of precise plateau ages on separate crystals reduces this span to ±0.5 Ma. A preliminary magnetostratigraphy (based on 41 flows from two sections) was briefly reported in [8], confirming the short time span. A revisit to these two sections in 1997 allowed us to increase the sampling density and to study some of the field relationships in more detail. The purpose of the present paper is to present the complete paleomagnetic data obtained in these sections and to complement the interpretation of [8] with new radiochronological and petro-geochemical data.

2. Sampling

Sixty-four paleomagnetic sites were sampled during the two field campaigns and initially labeled chronologically as PM1 to PM64. For the sake of clarity PM was transformed into LL or WT to allow for easy identification of the section (Lima-Limo or Wegel Tena) concerned. Nevertheless, the numbers were left unchanged with respect to [2,8]. The first and main section, called Lima-Limo (Figs. 1 and 2a), is composed of a succession of basaltic flows, 1950 m thick, with minor acidic tuff intercalations (Fig. 3a). The sequence of 43 flows and tuff units sampled in this section appears to include the very first and last flows of the trap series. Stratigraphic distance between sites varies from 5 to 110 m (Table 1), depending on outcrop quality, with an average of about 50 m. All sites (except LL24 and 25) are in a direct vertical sequence along an approximately N–S section of 15 km. Sites LL24 and 25 were sampled further to the NE at the bottom contact and at the contact between the Ashangi and Aiba Formations (see Section 4). From 6 to 10 cores were drilled at each site (except LL64 with a single oriented block) and oriented with both sun and magnetic compasses. One site corresponds to one cooling unit, with as regular a vertical distribution as possible. Because of poorer outcropping conditions at the bottom of the section, the sampling there was sometimes more patchy. Paleomagnetic results indicate that site LL21 probably corresponds to two cooling units, because the two sampling sub-sites (separated horizontally by 20 m and vertically by 3 m) yield very different directions, well-grouped for each sub-site.

The second section, 300 km to the SE (Fig. 1), on the road from Dessie to the village of Wegel Tena (Fig. 2b), is thinner (1280 m) and lacks the bottom contact (Fig. 3 and Table 2). It is interesting because it contains the Alaji acidic Formation, capped by a Termaber basaltic flow (WT37), and provides a test for lateral correlation within the trap province. In this section, 22 cooling units were sampled between 1770 and 3050 m, in two distinct sub-sections, below Wegel Tena for the higher part and below the Gishe Maryam monastery for the lower part. Correlation between the two subsections was easily performed using the prominent basaltic cliffs and acidic tuff sequences.

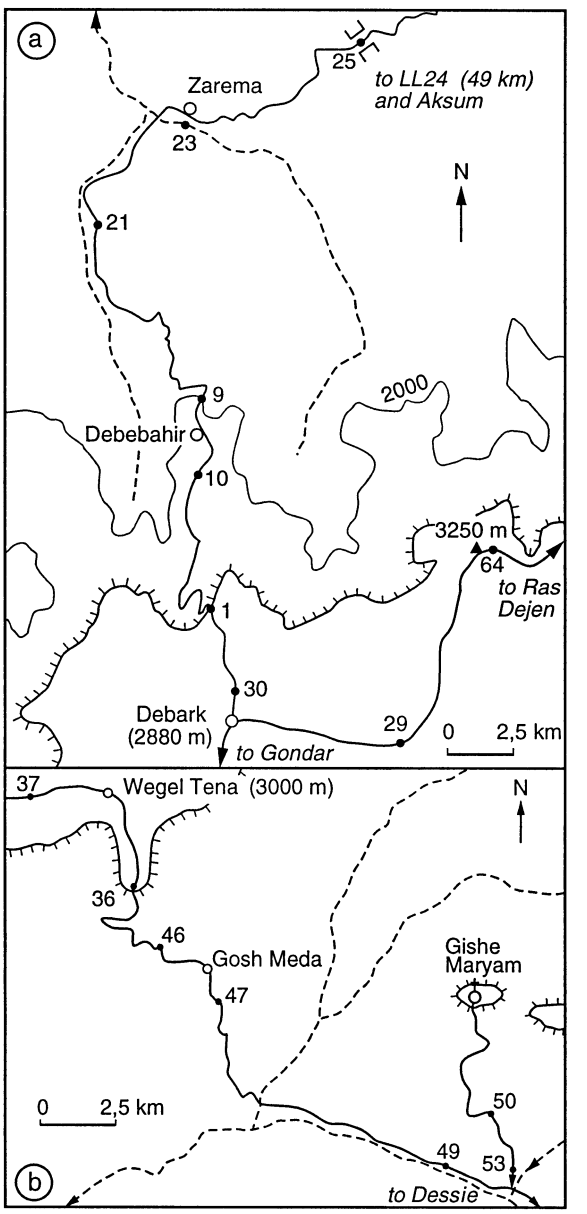


Fig. 2. Detailed map of the sampling in the two sections (a) Lima-Limo, (b) Wegel Tena (not all sites are indicated).

3. Analytical techniques

Natural remanent magnetization (NRM) was measured using a JR5 Agico spinner magnetometer in a shielded room. Stepwise thermal demagnetization was performed with a MMTD furnace, while

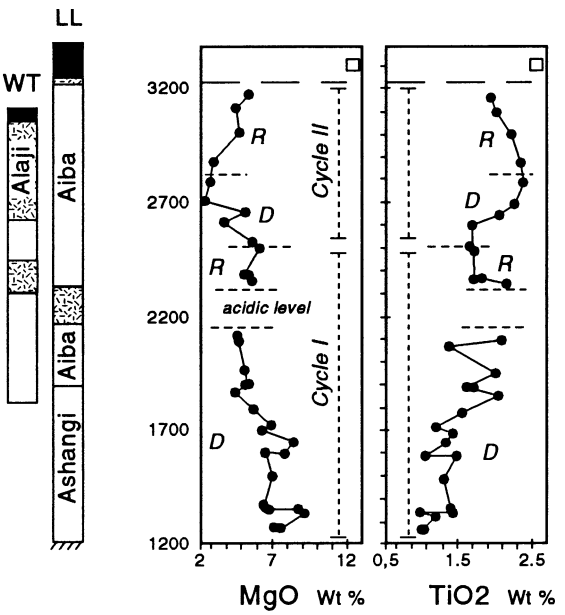


Fig. 3. Simplified stratigraphic sections of Lima-Limo (*LL*) and Wegel Tena (*WT*), showing lithology and geochemical logs (LL only) versus altitude. Basaltic and acidic trap products appear in white and gray, younger alkaline basalts of the Termaber formation in black, and underlying sedimentary rocks are stippled. In geochemical logs: full circles = traps, open square = Termaber, D = differentiation period, R = replenishment period.

Schonstedt and Molspin equipment were used for alternating field (AF) demagnetization. Two specimens from one pilot core per site were submitted to full progressive thermal and AF demagnetization. Demagnetization orthogonal plots were interpreted in terms of characteristic (ChRM) and secondary magnetizations using principal component analysis (PCA). Site mean results presented in Tables 1 and 2 are derived from PCA analysis of one to three specimens for all cores.

Thin sections from each cooling unit were examined by optical microscopy in order to select the least altered material for geochemistry and age determination. In these two sections, 16 samples were selected for the $^{40}Ar/^{39}Ar$ step-heating technique, including 3 new samples from acidic layers chosen specifically for the magnetostratigraphic correlation. Depending on grain size and K amount, either whole rock or bulk sample plagioclase for the basalts, or single feldspar grains for the acidic layers, were analyzed. A full description of the technique is given in Refs. [8,18]. The three new samples analyzed were dated on sanidine single grains which were irradiated in the McMaster reactor (McMaster University, Hamilton, Canada) with a total flux of 8.8×10^{18} n cm^{-2}, in the 5C position. The three samples were included into a narrow zone in order to minimize the effect of flux gradients, which is estimated to be on the order of $\pm 0.2\%$. The Hb3gr hornblende was used as a monitor (age = 1072 Ma; see references in Ref. [8]). Isotopic ratios were measured with a VG3600 mass spectrometer operating with a Daly system. Typical blank values were in the range 48–85, 0.7–10, $0.3 - 0.7 \times 10^{-14}$ cm^3 STP for masses 40, 39 and 36, respectively. Argon isotopes measured on the samples were on the order of $0.4 - 28 \times 10^2$, $0.34 - 240 \times 10^3$ and 2–11 times the blank level, respectively, for significant measurements. Thirty-five samples from the Lima-Limo section and 9 samples from the Wegel Tena section were analyzed for major elements by ICP–AES at the Universities of Aix-Marseille III and Clermont-Ferrand II.

4. Petrology and geochemical evolution

The Lima-Limo (LL) volcanic pile is predominantly composed of basaltic flows. Scarce acidic rocks (rhyolites) are present in the middle of the section, where they determine a flat geomorphological discontinuity (Fig. 3), and on the top of the plateau. In this section, basalts are transitional to tholeiitic and belong to the low-Ti magma type (LTi) identified in many CFB provinces [19,20] and recognized in Ethiopia [21]. Basalts from the Wegel Tena section all belong to the high-Ti (HTi) magma types [21] and are overlain by the Alaji acidic Formation. These basalts vary from olivine–clinopyroxene rich alkaline basalts to gabbroic assemblage bearing transitional basalts.

The Ashangi Formation (1/3 of the whole LL section, Fig. 3) is composed of thin lava flows (<10 m) and is characterized by a smooth morphology. By contrast, the Aiba Formation is a succession of massive cliffs corresponding to thick (10–50 m) basaltic flows. Basalts from Ashangi have intergranular to intersertal textures often exhibiting a sub-ophitic tendency. They display the most primary compositions

Table 1
Rock magnetic and paleomagnetic results for Lima-Limo section (13.25ºN, 37.93ºE)

ID	Altitude (m)	N/n	Dec (º)	Inc (º)	k	α_{95}	NRM (A/m)	MDF (mT)	T_b (ºC)	K_m (10^{-3})
LL24	1275	9/7	191.4	−8.8	88	6.5	1.1	31	480	16.2
LL23	1330	10/6	199.4	34.4	149	5.5	4.8	30	450	15.7
LL22	1360	7/7	193.6	37.9	197	4.3	3.4	29	530	13.2
LL21A	1395	7/7	350.1	−20	188	5.6	3.3	16	480	15.7
LL21B	1390	3/3	168.1	4.5	1000	2.7	2.3	15	480	23.3
LL20	1500	7/5	191.1	−38.6	91	8.1	2.5	21	490	34.6
LL19	1580	7/6	187.2	−22.5	79	7.6	2.2	21	530	66
LL18	1645	10/7	154	−8.7	12	18.1	3.3	13	450	47.3
LL17	1690	8/7	190.4	−9.2	153	4.9	1.6	29	530	24.9
LL16	1725	10/8	184.4	25.7	291	3.3	2.6	35	505	34.4
LL15	1760	9/9	203.7	33.1	32	9.3	0.87	19	480	84
LL14	1790	7/6	173.5	−21.7	95	6.9	2.4	15	530	57
LL54	1835	11/11	182.1	10.4	97	4.6	2.4	33	575	28.2
LL25	1845	9/5	75.1	46.3	48	11.2	1.7	20	530	25.5
LL55	1865	9/9	192.6	5.2	18	12.4	1.4	13	300	89
LL56	1885	9/9	2.6	4.7	83	5.7	1.7	>100	400	61
LL13	1900	7/5	349.8	−7.4	268	4.7	1.6	14	505	48
LL12	1960	9/6	8.9	−19.2	41	10.6	43	13	375	55
LL57	2015	5/4	9	19.9	564	3.9	3.1	7	500	71
LL11	2030	10/8	343.8	40.2	102	5.5	4.1	29	500	47.5
LL58	2080	14/14	28	−30.8	39	6.4	2.4	26	600	34
LL09	2110	5/4	359.6	3.7	203	6.9	6.1	23	500	62
LL61[a]	2120	5/5	342.5	−19.7	54	10.5	0.39	>100	500	5.3
LL60	2185	8/8	10.2	22.3	58	7.3	0.65	12	450	32
LL59	2235	6/6	356.8	−8	131	5.9	2.1	38	550	35.6
LL10	2330	8/5	188.9	8	219	5.2	0.96	46	500	38.9
LL62	2360	5/4	190.4	7.9	127	8.2	3.6	>100	400	32
LL08	2462	9/7	175.6	−18.6	382	3.1	2.4	46	505	12.7
LL07	2500	11/6	188.1	19.1	278	4	1.9	40	490	13.5
LL06	2510	12/6	176.6	−33.2	284	4	9.3	25	475	38.3
LL05	2555	9/8	218.5	41.2	214	3.8	0.99	>100	555	5.1
LL04	2615	9/7	172.5	0	507	2.7	1.8	60	530	20
LL03	2660	10/8	176.1	0.6	71	6.6	1.2	28	490	14.9
LL63	2710	11/9	204.5	−31.8	57	6.9	9.4	16	300	12.9
LL02	2790	10/7	178.8	16.8	57	8.1	0.93	14	500	16.2
LL01	2885	9/8	171.1	21.5	84	6.1	1.9	19	530	15.7
LL30	2900	7/5	175	21.6	253	4.8	1.2	26	480	19.8
LL29	3015	8/5	182.1	20	72	9.1	22	14	480	27.2
LL26	3060	10/9	179.2	−15.5	222	3.5	2.0	40	505	20.7
LL27	3115	9/7	176.2	−12.3	456	2.8	2.2	41	505	8.5
LL28	3165	9/6	173.2	−6	455	3.1	0.96	35	505	11.4
LL64[a]	3225	5/5	207.3	3.7	114	7.2	0.42	12	500	7.4
Mean normal flows (excl. 25)		11	0.1	−1.6	10.5	14.8				
Mean reverse flows (excl. 5, 25)		29	183.4	1.6	11.7	8.2				
All LL flows (excl. 5, 25)		40	2.5	−1.6	11.7	6.9				

[a] Indicates acidic flows; N and n are the numbers of specimens measured and used in the mean, respectively; various mean directions are shown at the end of the table.

of the whole section (5% < MgO < 9%). The Aiba Formation is composed of more differentiated basalts (2% < MgO < 6%) with glomerophyric, rarely aphyric, textures. According to petrology, the Lima-Limo section exhibits an evolution towards more differentiated terms in the upper part of the

Table 2
Same as Table 1 for Wegel Tena section (11.5ºN, 39.25ºE)

ID	Altitude (m)	N/n	Dec (º)	Inc (º)	k	α_{95}	NRM (A/m)	MDF (mT)	T_b (ºC)	K_m (10^{-3})
WT49	1770	20/20	324.7	−0.2	53	4.5	2.8	15	550	40
WT53	1870	8/8	357.5	0.7	84	6.1	2.1	7	300	27
WT52	1970	7/5	6.2	6.9	191	4.9	3.3	16	400	78
WT51 [a]	2090	11/11	346.9	13.8	3.9	26.4	0.007	15	300	0.12
WT50	2190	11/11	7.2	18.1	53	4.5	3.5	16	550	51
WT47	2235	9/9	358.9	−14.7	206	3.6	4.6	17	550	46
WT48	2350	13/13	169	−14.2	344	2.2	16.5	>100	525	6.4
WT46	2410	15/13	186.7	−20.1	66	5.1	21.5	35	500	14
WT40	2430	9/8	183.7	−27.2	185	4.1	11.4	58	517	13
WT45	2500	11/10	188.8	−7.6	59	6.4	7.9	35	500	28
WT39	2535	10/8	180.5	−6.1	255	3.5	14.8	37	480	7.2
WT31 [a]	2600	15/15	168.3	64	91	5.4	0.25	25	580	16.1
WT44 [a]	2625	14/14	184.3	−21.1	126	3.6	0.75	20	500	13.9
WT32 [a]	2660	9/8	174.4	−34	121	5	0.38	23	530	0.74
WT43 [a]	2675	10/9	348.6	2.6	12.6	15.1	0.001	19	300	0.15
WT33 [a]	2680	6/4	6.6	−6.3	68	11.2	0.46	>100	530	2.2
WT34 [a]	2765	8/8	6.4	0.9	105	5.4	0.43	15	600	7.3
WT42 [a]	2780	6/6	354.3	0.1	9	23.6	0.0004	35	300	0.1
WT38 [a]	2797	13/13	177.8	28.3	153	3.4	1.99	>100	600	22.4
WT35 [a]	2900	9/8	190.5	46.5	44	8.5	0.04	27	580	1.4
WT36 [a]	2940	9/9	337.3	18.5	3.2	34	8.5	15	300	11
WT41 [a]	2930	13/12	337.9	−4.6	185	3.2	0.54	50	550	20
WT37	3050	7/4	354.1	26.7	39	15	16.8	8	400	5.8
Mean normal flows (excl. 36,49)		11	356.8	4	28.8	8.7				
Mean reverse flows (excl. 31)		9	181.6	−7.1	9.1	18				
All WT flows (excl. 31, 36, 49)		20	358.9	5.4	15.2	8.6				

[a] Indicates acidic flows; N and n are the numbers of specimens measured and used in the mean, respectively; various mean directions are shown at the end of the table.

lava pile. However, at a smaller scale, this general evolution shows more complex variations (Fig. 3). These variations are controlled either by fractionation of a gabbroic assemblage (olivine + plagioclase ± pyroxene; [21]) or by (possibly periodic) replenishments of the reservoir by new magma. During fractionation periods (D in Fig. 3), incompatible element content (TiO_2) increases sharply as MgO content decreases. Conversely, short periods of increasing MgO and decreasing incompatible element (R in Fig. 3) are thought to reflect mixing in the crustal magma chambers, between residual resident material and new input of primary melts [21]. Two main magmatic cycles (differentiation + replenishment) have been distinguished in the geochemical evolution of this basaltic pile. The first and most important volcanic cycle was punctuated by the emission of rhyolites between the differentiation and the replenishment events. Note that trap emission ended during a replenishment period (second cycle), suggesting that a huge quantity of magma was still trapped in the crust. The overlying basalts belong to the alkaline, central-vent related, Termaber Formation (20.5–22 Ma, [2]), and display contrasting alkaline compositions compared to the traps (Fig. 3).

This coherent magmatic evolution, highlighted by two well identified volcanic cycles, strongly suggests the existence of a large magmatic storage system rather than smaller disconnected reservoirs. In the latter case, erratic geochemical variations would have been expected. Moreover, the huge volumes of magma erupted during the building of this volcanic pile also require the presence of such large storage systems.

5. Paleomagnetic analysis

5.1. NRM analysis

In the basaltic sites, average NRM intensities and magnetic susceptibilities (K) vary in the range 0.6–41 A/m and 5–90 $\times$ 10^{-3} SI, respectively. Apart from a few very 'hard' sites, median destructive fields (MDF) of NRM are less than 40 mT and maximum unblocking temperatures are between 300 and 580ºC. Such values are typical of a few % of more or less titanium substituted and oxidized magnetites. The stronger MDF found in some cases may be the sign of single domain grain size, strong oxidation of the magnetite grain, or alternately could be explained by some titanohematite. Thin-section evidence of high temperature deuteritic oxidation often accompanies an increase in coercivity. The acidic layers show lower NRM and K ranges: 0.001–2 A/m and 0.1–7 $\times$ 10^{-3} SI, respectively. MDF values are not larger than values encountered in the basalts, but maximum unblocking temperatures are usually higher, up to 620ºC. These features may correspond to smaller amounts of iron oxides of spinel type, or to a more pronounced presence of titanohematite.

High intensity, highly scattered directions and rapid decay of NRM under AF allowed us to identify 3 sites struck by lightning (LL12, WT36, WT37). Tentative ChRM directions are reported from 50 to 100 mT PCA analysis on selected samples. The vast majority of unstruck sites show very simple demagnetization diagrams, with no or quite limited secondary magnetization, easily erased by 10 to 30 mT, or 100 to 300ºC. In the usual case when AF and thermal demagnetization of pilot specimens yield coherent Zijderveld plots with unidirectional decay to the origin, AF demagnetization was chosen for the full site demagnetization. In a few cases, AF demagnetization was found to be insufficient and thermal demagnetization was preferred to obtain the site-mean ChRM.

Results of the Fisher statistics applied to ChRM directions reveal well defined mean directions, with α_{95} less than 8º in 49 out of 65 flows. Only seven sites (LL18, LL55, WT51, WT43, WT42, WT36, WT37) show poorly defined directions with $K < 30$ and $\alpha_{95} > 12$º, among which two were struck by lightning (WT 36, WT 37). The other dispersed sites, except LL18 and LL55, correspond to very weakly magnetized tuffs of the WT section, where alteration may have played a role. Some struck sites show good grouping after full AF demagnetization and exclusion of some outliers (e.g. LL12). The efficiency of AF demagnetization is further attested by the coherency between two sites from the same flow (WT36, WT41), the first one only having been struck. Visual inspection of the outcrops, as well as the continuity of exposed cliffs over tens of km in the landscape, always indicate horizontality, so that no tectonic correction was applied.

5.2. Paleosecular variation (PSV) and virtual geomagnetic pole (VGP) positions

Intermediate directions are better dealt with following Vandamme [22], who uses a recursively estimated cutoff angle between mean and individual VGP instead of the usual a priori cutoff angle of 40º [23]. This leads to the identification of sites 5 and 25 in Lima-Limo and sites 31 and 49 in Wegel Tena as excursional directions (Fig. 4a). The optimal cutoff angle is found to be 30.6º, which excludes one more site compared to the 40º a priori angle. A lower cutoff angle (26º) was found in the 0–2 Ma equatorial lavas of Galapagos islands [24] linked to much smaller dispersion. Mean normal and reverse directions for each section are not significantly different, suggesting a good averaging of PSV and proper identification of ChRM (Tables 1 and 2). Moreover, the LL and WT poles are not significantly different (7.5º $\pm$ 9.3º), so that we prefer to consider directly the combined LL–WT mean VGP and its angular standard deviation (ASD). We excluded the poorly defined (Section 5.1) and excursional sites, together with site LL64 (defined only by one block) from the mean considered to best characterize the trap series. The ASD, corrected for within-site dispersion following [23], is 14.9º (with a 13.1–17.1º range). This is not significantly different from the global average extrapolated at the equator for the 22.5–45 Ma period: 15.4º [23].

Comparison of our mean pole can be made with mean VGPs proposed in earlier studies of the Ethiopian traps. These studies were however not focused on the trap series itself and have poorer

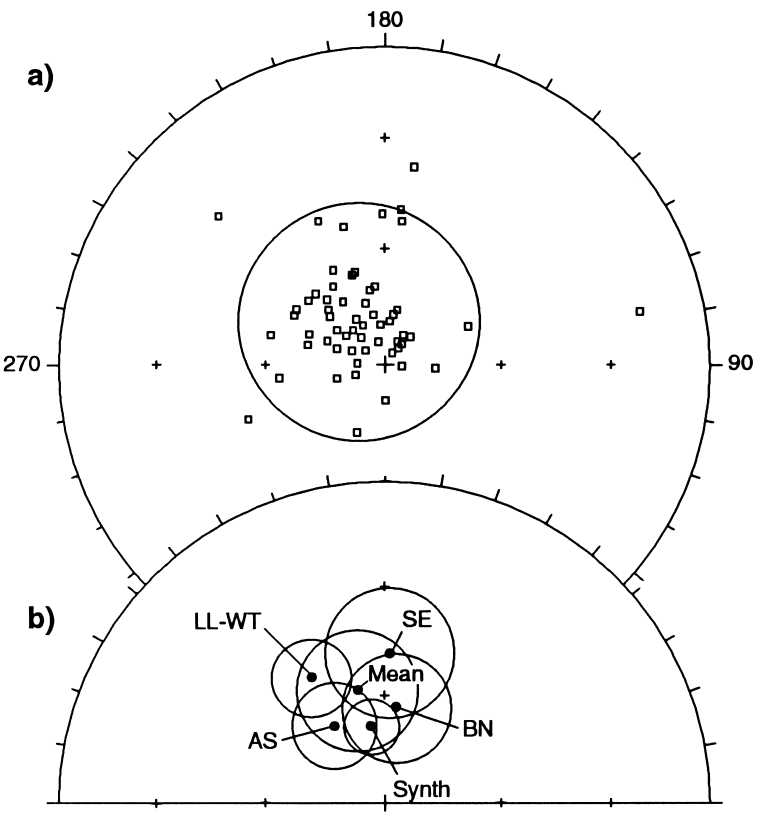


Fig. 4. (a) Stereoplot of VGP directions (all sites with $K > 30$) for the two sections, with the circle defining non-transitional directions following [22]; (b) mean VGP directions obtained in this study compared to 30 Ma synthetic pole for Africa of Ref. [26], together with selected poles for the traps in Ethiopia and Arabia (see Table 3 and text). Last plot is restricted to latitude $>60^{o}$.

radiochronological and stratigraphic control; it is likely that Termaber or even younger volcanics were included. The Brock et al. study [15] is based on two hand-samples per flow, and therefore has limited directional reliability. It corresponds to three stratigraphic sections among which the third is probably from the Miocene Termaber Formation, while the others correspond to the Blue Nile section (BN in Fig. 1). The Schult study [16] in the SE Plateau (east side of the rift, see Fig. 1) is better constrained. These results (Table 3) are based on several cores per site and blanket AF demagnetization following full demagnetization of pilot samples. A more recent study concerns the Saudi Arabia satellite of the Yemen traps, the As Sarat volcanics, with a base dated by K/Ar at 29.4 ± 2 Ma [17]; the As Sarat data should be rotated by 8° clockwise [25] to account for post-trap breakup of Arabia away from Africa.

Table 3
Mean poles obtained for the Ethiopian traps in this study (LL–WT), as well as previous studies (Fig. 1 and text) compared to reference poles for Africa at 30 Ma

Site	N	Lat.	Long.	A_{95}
LL–WT	53	77	208	3.7
AS [17]	42	79	248	4
Rotated back	–	82	208	–
SE [16]	22	75	170	6
Rotated back	–	76	175	–
BN [15]	20	81	168	5
Trap mean	4	79	189	5.6
Synth [26]	32	83	183	2.5
NDF [28]	4	76	193	2.7

Consistency between the BN, SE, AS poles and ours (LL–WT) is reasonable (Fig. 4b, Table 3). The overall mean is not statistically different from the 30

Ma synthetic pole for Africa (20 Myr window), based on a review of 32 carefully selected data [26]. Yet, it is puzzling that our pole alone, which might be considered as the best constrained for the Ethiopian traps, is itself the most remote from the synthetic pole. There is no relative rotation implied, but a latitudinal displacement, with the LL–WT pole being far-sided by $6^{o} \pm 6^{o}$, possibly explained by a steady quadrupole term of 14%. For comparison a steady quadrupole term of 5% amplitude with respect to the dipole is necessary to account for the far-sidedness found in the 0–5 Ma period [27], while a rather large non-dipole field has been evidenced in inclination data from Oligocene sediments around the African plate [28]. Indeed, the 30 Ma African pole derived in [28] agrees perfectly with our pole (Table 3).

6. Magnetostratigraphy and $^{40}Ar/^{39}Ar$ ages

6.1. Lima-Limo section

The 1950 m Lima-Limo section yields a sequence of three chrons, respectively of 600, 410 and 940 m thickness (Fig. 5), with only the central normal chron completely recorded. The apparently normal polarity found in the upper part of site LL21, in the middle of the older reverse chron, may be interpreted either as an excursion or as a rotated block, due to the very limited extent of the outcrop (see Section 2). The lack of large (i.e. >0.1 Myr) hiatuses within the section is attested to by the lack of observed intraflow paleosoils or sediments, except for some baked clay layers of a few tens of cm, and by the coherent petrogenetic evolution. Whichever correlation is made with the reversal timescale around 30 Ma, it confirms that the whole section was emplaced in less than 2 Myr. This time span corresponds to a minimum emplacement rate on the order of 1 m/kyr, or an average interval of 50 kyr between flows. The estimated duration of a reversal being on the order of 5 kyr, it is not surprising that only one intermediate direction was found, sandwiched between two flows with opposite polarities, while the three others (to which LL21N can be added) should correspond to within-chron excursions. Unfortunately this transitional direction (LL25) does not appear within the main Lima-Limo sequence, but in a nearby section (Fig. 2a). We attempted to resample this intermediate direction between sites LL13 and LL14, but no intermediate directions were found in the new sites (LL54 to 56). This places an upper limit of 20 m on the transitional zone thickness, which is less than 20 kyr according to the emplacement rate estimated for the whole section. Therefore, the total estimate, based on chron duration, as well as the local estimate, based on reversal duration, are consistent.

The argument for a regular emplacement rate throughout the sequence may be thought questionable because of the coincidences between lithological changes and reversals. The reverse to normal transition coincides approximately with the Ashangi to Aiba transition marked in both flow thickness and morphology. However, a significant hiatus at this reversal can apparently be excluded on the basis of field and petrogeochemical evidence (Section 4), as well as the LL25 record of a transitional direction. On the other hand, the next N-R reversal in Lima-Limo occurs at the top of the 200-m-thick succession of acidic material, hydrothermalized basalts and tuffs, marked by a morphological break corresponding to a strong lithological contrast. Geochemical analysis has shown that this level corresponds to the end of magma chamber purging and to a relatively quiescent period. The overlying basaltic sequence, corresponding to the second period of active magma feeding is entirely reversed. It is therefore possible that part of the geomagnetic field record is missing at this contact [8], although careful search of continuous outcrops around Debebahir did not show any developed soil or sediment layer.

The already published $^{40}Ar/^{39}Ar$ ages [8] are summarized in Fig. 5. Three new high-precision ages have been obtained from acidic units in Lima-Limo (LLA, LLB and LLC, at 2120, 2250 and 3220 m, respectively, Fig. 6). Two LLA sanidines display similar plateau ages of 30.1 ± 0.1 Ma, concordant with the high temperature apparent ages displayed by the two other grains of the same sample. Although one of these last two grains also gave a plateau age following the previously defined criteria, their low temperature apparent ages are affected by alteration phases, as shown by the higher $^{37}Ar_{Ca}/^{39}Ar_{K}$ ratios. Therefore, the best estimate of the age is given by the two plateau ages. The two grains of LLB sample displayed plateau ages at 30.2 ± 0.1 and

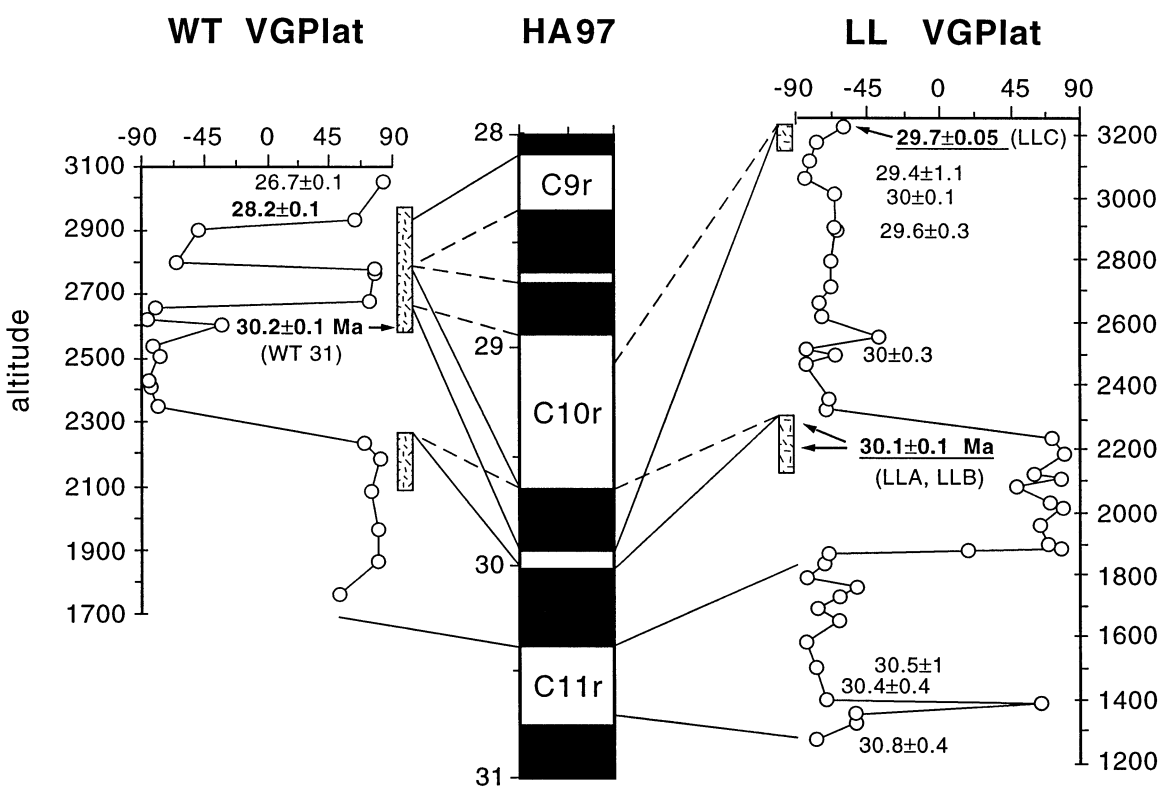


Fig. 5. Vertical log of paleomagnetic polarity (shown by VGP latitude) in the Lima-Limo (*LL*) and Wegel Tena (*WT*) sections, with $^{40}Ar/^{39}Ar$ ages, correlated with the HA97 reversal timescale [29]. Only well defined plateau ages (or other robust ages discussed in [8]) have been selected, with data from rhyolitic sanidines appearing in bold (new data underlined). Gray blocks corresponds to the acidic or mixed levels.

29.7±0.2 Ma, concordant at the 2 sigma level, giving a weighted mean of 30.1 ± 0.1 Ma, similar to the LLA plateau age. Significantly younger ages were obtained on the 3 grains of the LLC sample, which displayed 2 concordant plateau ages of 29.70 ± 0.05 and 29.65 ± 0.06 Ma, similar to the fusion step apparent age of the third grain.

The reference reversal timescale of Fig. 5 is HA97 [29], a revision of the CK95 timescale [30], which uses the same calibration dates but gives equal weight to the various magnetic profiles instead of choosing a 'master' South Atlantic profile as done in Ref. [30]. HA97 is shifted by 0.2 to 0.3 Ma toward older ages with respect to CK95 in our study interval. The most satisfactory correlation in terms of emplacement rate regularity is that the section begins in Chron C11r and ends in Chron C10r (dashed lines in Fig. 5). This leads to a maximum time interval between 28.9 and 30.8 Ma. This 1.9 Ma maximum duration would correspond to a mean emplacement rate >1.0 m/kyr. The respective rates for the successive polarities from bottom to top would be >1.5, 0.6 and >1.2 m/kyr. This interpretation assumes that the subchron within Chron C11n, lasting about 105 kyr, has been missed around the altitude of 2000 m, or corresponds to the hypothetical hiatus around 2300 m. In the latter hypothesis it implies that the upper part of normal Chron C11n is also missing. The total duration of the hiatus would then be at least 400 kyr, increasing the average emplacement rate outside the hiatus to about 1.5 m/kyr. An alternative correlation, placing more emphasis on the new $^{40}Ar/^{39}Ar$ ages, particularly the top LLC age, would link the reverse subchron within Chron 11n to the entire upper reversed section (full lines in Fig. 5). This would lead to an emplacement rate sequence >1.5, 1.2 and >8.5 m/kyr — with total duration less than 800 kyr. Geochemical evidence, as well

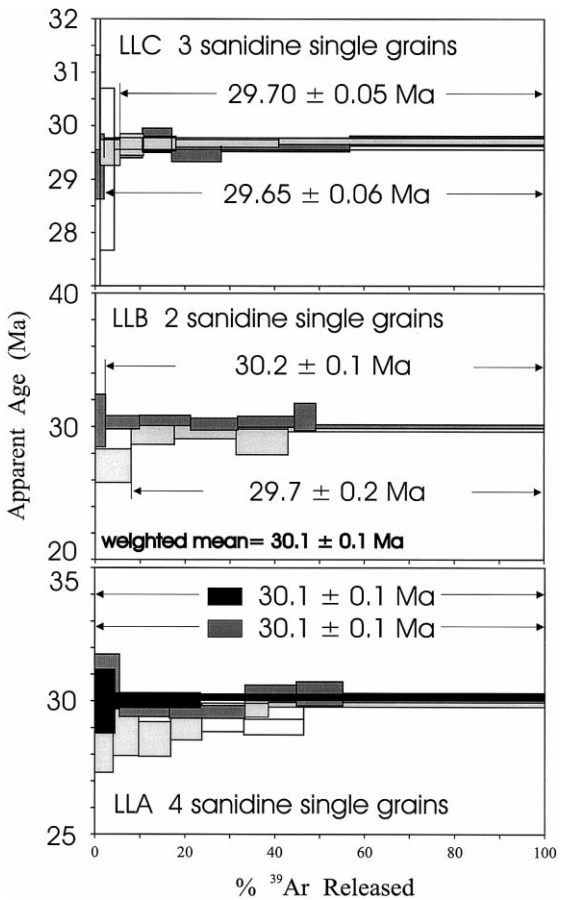


Fig. 6. New $^{40}Ar/^{39}Ar$ age spectra obtained on 9 sanidine single grains from 3 acidic tuff or flows in Lima-Limo section (full data available as an **EPSL Online Background Dataset** http://www.elsevier.nl/locate/epsl, mirror site: http://www.elsevier.com/locate/epsl). All errors on individual apparent ages are quoted at the one sigma level and do not include the error on the $^{40}Ar^{*}/^{39}Ar_{K}$ ratio and age of the monitor. The error on the $^{40}Ar^{*}/^{39}Ar_{K}$ ratio of the monitor is included in the calculation of the plateau age error bars (given at the one sigma level). The criteria given in [8] were used to define a plateau age.

as the exceptional thickness (near 100 m) in several flows of the upper reverse section, make such an acceleration of emplacement rate plausible. In that interpretation the beginning of subchron C11n.1r would be younger than 30.1 ± 0.1 (LLA, LLB) and its end younger than 29.7±0.05 (LLC). The absolute ages of C11n.1r and C11r given by CK95 are 29.65–29.75 and 30.1–30.5 Ma, while they are 29.9–30 Ma and 30.4–30.8 Ma for HA97, respectively. Neither scale is fully consistent with respect to our analytical errors, but we favor HA97, because LLA and LLB should appear at the end of Chron C11r in CK95.

6.2. Wegel Tena section

This 1280-m-thick section yields two more chrons than the Lima-Limo section, but only two are recorded below the Alaji Formation. Using petrology, $^{40}Ar/^{39}Ar$ ages, thickness and altitude, it is tempting to correlate the lower normal chron (500 m thick) to the normal chron in Lima-Limo (410 m thick), both of which end with mixed basic and acidic products near 2300 m altitude. On the other hand, the upper basaltic section would be much thinner in Wegel Tena than in Lima-Limo (300 vs. 900 m), with the topmost 400 m of the Alaji Formation postdating the end of the Lima-Limo section. This would agree with younger ages found on top of Alaji Formation (WT35 and WT37 at 28.2 and 26.7 Ma respectively) and suggests that the top normal chron is either Chron C10n or Chron C9n. The middle normal Chron (around 2700 m within the Alaji Formation) would be either Chron C10n or the upper part of Chron C11n, in agreement with the second possibility discussed for Lima-Limo. In these two cases the sequence of emplacement rates would be >0.7, 0.5, 0.2, 0.4 m/kyr and >1.5, 3.6, 0.4, 0.2 m/kyr, respectively. The estimate for the last chron is not mentioned because it includes the long hiatus between the Alaji and Termaber Formations. In both solutions a prominent decrease of emplacement rate occurs at the end of emplacement. The age of WT31 (30.2 ± 0.1 Ma) excludes its correlation with Chron C10r but is a bit old for subchron C11n.1r. In the CK95 scale it may even be assigned to Chron C11r. However, this last solution is at odds with stratigraphic evidence which leans toward a correlation of WT31 with LLC. Since we have more confidence on the numerous and concordant ages of the LL section, we believe that the solution proposed in Fig. 5 (full lines) is the most robust.

6.3. Extrapolation to the whole Ethiopian province (age and duration)

The onset of volcanism took place between 29.5 and 30.5 Ma all around the province, from the Blue

Nile [2] section (basal $^{40}Ar/^{39}Ar$ age at 29.4 ± 0.3 Ma) to As Sarat [17], including Adigrat [8], Eritrea [31] and Yemen [32,33]. This onset occurs accordingly in Chron C11r (30.4–30.8 Ma following [29]) in the LL paleomagnetic section. Detailed stratigraphic studies in Yemen [33] and Ethiopia (this study) suggest that the major part of the sequence was emplaced in about 1 Myr or less. Younger ages (28–26 Ma) are found in the topmost formations and may correspond to minor volcanic pulses, clearly separate from the major flood event. This interpretation is further supported by the magnetostratigraphic record of only 3 chrons for this major pulse. Interestingly, the As Sarat basaltic section, 500-m-thick and starting on the pre-trap basement, shows 300 m of thin reversed flows, overlain by thick normal flows. So the polarity and lithological sequences are the same as for LL, while the thinning of the reverse chron by a factor of two could indicate a thinner or slightly younger pile in As Sarat. The only other paleomagnetic study of interest for magnetostratigraphy concerns the Blue Nile section and another nearby [12], both of about 200 m thickness and starting at the basement contact. These two sections are entirely normal and could be correlated to the base of the WT section and the middle of LL section. Extrapolating safely the LL section results to the whole Ethiopian province would of course require more sections to be investigated. However the above discussion, together with the fact that the LL section is the thickest and most complete known in the province, also in a rather central position (Fig. 1), argues for a possible use of our Lima-Limo results as a stratotype for the whole province. The duration of the main pulse of flood volcanism throughout the province may thus be estimated as less than 0.8 or 1.9 Myr depending on the magnetostratigraphic interpretation.

7. Climatic impact of Ethiopian trap emplacement

The volume of the province usually reported [12,33], 350,000 km^3, deserves some discussion. The present surface (600,000 km^2 [12]) is not consistent with this volume, as it would lead to an average thickness of 600 m. This is well below the reported thickness of most exposed sections, usually between 1 and 2 km. The above surface estimate includes the Southern Ethiopian province where flood volcanism may be somewhat older [13], therefore possibly not synchronous with the main Ethiopian traps. Subtracting their surface reduces the present trap surface to about 400,000 km^2. However, this estimate neglects the effect of erosion which, due to the large altitude contrast and the dissection of the plateau edges, must have reduced the initial surface by a significant amount [34]. Uplift of the Red Sea shoulders would also have increased the efficiency of erosion, as indicated by the small size of trap remnants in Eritrea and Adigrat, where the basement top is now at 2700 m altitude. A simple lower estimate of initial trap volume can be obtained by drawing a circle (Fig. 1), enclosing most remnants of the traps on a reconstruction of the region, after closure of the Red Sea and Gulf of Aden [25,35]. This circle cannot have a radius less than 500 km, thus leading to a minimal surface of 800,000 km^2. Taking an average thickness of 1.5 km, the initial minimum volume should have been 1.2×10^6 km^3, leading to a minimum average magma eruption rate of 0.6 or 1.6 km^3/yr, depending on the magnetostratigraphic solution. This flux is typical of other major flood basalt provinces and has the required order of magnitude to suspect that trap emplacement had a major climatic impact [1,5]. The efficiency of this impact may have been greatly enhanced by the presence of acidic products (on the order of 5–10% of total volume) whose mode of explosive activity is better able to disperse aerosols at stratospheric altitudes. Evidence for individual pulses in the emplacement of the Lima-Limo section, and particularly the faster magnetostratigraphic solution, point to two discrete periods with much larger flux (within the total duration of trap emplacement).

A recent synthesis [36] argues that a major global cooling event (Oi2), marked by a sudden δO^{18} shift, generalized emersions of Atlantic Ocean margins, and major development of the Antarctic ice sheet, occurred at the base of Chron C11r. The coincidence, proposed in Ref. [8], between Ethiopian trap emplacement and this major cooling event is clearly striking. However, this correlation could be disputed, based on uncertainties in the calibration points of the reference scale. Confirmation could

come from finding trap-related products within well dated marine sequences. The major lower Oligocene tephra layers found in leg 115 [37], 2600 km to the Southeast in the Indian Ocean, could be such a trap-related by-product. This multiple acidic tephra layer, found in all leg 115 sites, is distributed over 0.35 Myr according to magnetic susceptibility curves and biostratigraphy, and contains the boundary between biozones NP23 and NP24 [38]. This limit is placed at the base of Chron C11n [39], that is at the time of the first acidic layers found in the LL and WT volcanic piles. Moreover, a rather sharp drop in δO^{18} of the fine carbonate fraction coincides with the leg 115 tephra layer [40]. We tentatively correlate this isotopic event to Oi2.

8. Conclusions

The detailed magnetostratigraphic, radiochronological and petrogeochemical study of a 2-km-thick complete reference section of the Ethiopian traps at Lima-Limo indicates that the section was emplaced within less than 1.9 (possibly including a 0.4 Myr hiatus) or 0.8 Myr during Chron 11, i.e. at about 30 Ma. The shorter solution seems to be favored by the $^{40}Ar/^{39}Ar$ ages. According to petrogeochemistry and geomorphology, emplacement rates probably reached maximum values within two distinct, discrete periods, separated by a halt in magma eruptions (and prior to that, chamber filling). Through correlation with the Wegel Tena section and a synthesis of other published data, we propose to extend this timing to the whole Ethiopian province, possibly excluding its southern Eocene part. We propose a trap duration much reduced compared to previous estimates because stratigraphic study has allowed the main tholeiitic flood volcanic sequence to be separated from subsequent minor alkaline volcanic pulses, and also because more precise ages could be measured. Reevaluation of initial trap volume to more than 10^6 km^3 indicates an average magma flux on the order of 1 km^3/yr (possibly significantly larger during the second active episode), typical of most large igneous provinces and sufficient to induce global climatic cooling. Trap age coincides with a major δO^{18} event (Oi2), and a global sea level drop of 40–80 m. Our work therefore confirms the probable global climatic impact of Ethiopian trap emplacement [8]. Finer timing of volcanic episodes can be expected from ongoing analysis of contemporary tephra-bearing ODP sediments.

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